

Macroecological and macroevolutionary perspectives on the Cenozoic history of South American Mammals

Lucas Buffan

Table of contents

Foreword and Acknowledgements	i
Foreword	i
Acknowledgements	i
1 The unevenness of biodiversity: corner stone for the large-scale study of life	1
1.1 The heterogeneous organisation of life across space	1
1.2 From peaks to declines: temporally-heterogeneous biodiversity	2
1.3 Inter-clade heterogeneity	5
1.4 Methodological biases arising when studying biodiversity	6
1.4.1 Biodiversity shortfalls	6
1.4.2 Propagation of the caveats of the fossil record when studying biodiversity through time	6
2 Disentangling biodiversity patterns across space and time: the macro-eco-evolutionary perspective	8
2.1 Context and terminology	8
2.2 Macroevolution: evolution above the species level	8
2.2.1 Tempo of Macroevolution: how do rates change across time?	8
2.2.2 Mode of Macroevolution: what drives macroevolutionary rate variations?	9
2.3 Macro-eco-evolutionary understanding of the spatial organisation of life: insights from the Latitudinal Biodiversity Gradient	10
3 Analytical tools to study large-scale biodiversity patterns	12
3.1 Macroevolutionary approaches to study how clades thrive and go extinct	12
3.1.1 Phylogeny-based	12
3.1.2 Fossil occurrence-based	12
3.1.3 Towards integrating multiple sources of data	12
3.2 Integrating ecological processes with evolutionary forces: mechanistic eco-evolutionary simulations	12
4 A strange biodiversity on a strange land : the case of mammals in South America	13
4.1 South America across the Cenozoic: geographical and environmental context	13
4.1.1 Amazonia: a unique and dynamic tropical ecoregion	13
4.1.2 The rise of the Andes : emergence of one of the largest and highest mountain chains, and its consequences on biodiversity	13
4.1.3 Intermediate flooding events	13
4.1.4 South America as a playground for evolutionary biologists	13
4.2 ‘Splendid isolation’: the captivating history of South American mammals across the Cenozoic	13
4.2.1 History of mammal palaeontology in South America: a brief overview	14
4.2.2 First stratum: endemic radiations (late Cretaceous–middle Eocene)	14

4.2.3	Second stratum: Palaeogene migrant arrival and diversification (late Eocene–Late Miocene)	15
4.2.4	Third stratum: North American migrant arrival, massive faunal disruption, and establishment of the extant mammal biodiversity across the Neotropics (latest Miocene–Present)	15
4.3	Challenges for South American mammal palaeontology	15
5	Objectives of this doctoral work	16
5.1	Deciphering the dynamics of diversification of South American mammals and their likely drivers across the Cenozoic	16
5.1.1	Elucidating the impact of the Eocene–Oligocene Transition on the macroevolutionary dynamics of South American mammals	16
5.1.2	Teasing apart the legacy of the Neogene continent-scale environmental upheavals on the macroevolution of SAM	16
5.1.3	Investigating what drove the emergence and demise of the largest rodents ever fooling the Earth: Chinchilloidea	16
5.2	Assessing the role of palaeoclimate in the emergence of the asymmetry behind the Great American Biotic Interchange	16
Chapter 1.	Deciphering the trends and drivers of South American mammal large-scale evolution across the Cenozoic	17
Chapter 2.1.	The fate of South America’s endemic mammalian fauna in response to the most dramatic Cenozoic climate disruption	17
Chapter 2.2.	Continent-scale insights into the last 23 million years of South American mammalian evolution	17
Chapter 2.	A matter of size: what underpinned the rise and demise of the largest rodents that ever lived?	18
Chapter 3.	Palaeoclimate alone fails to explain the asymmetry of the Great American Biotic Interchange	19
Discussion		20
Limitations		20
Terminology, definitions and other macroevolutionary enigmas		20
Perspectives		20
An almost closed system: founding migrations in the Palaeogene history of South American mammals		20
Sustainable late Cenozoic connection to North America: what drove the asymmetry of the GABI?		20
Future directions towards an improved understanding of the emergence and dynamics of South American ecosystems		21
Concluding remarks		22
Glossary		23
Literature cited		25

Foreword and Acknowledgements

Foreword

Acknowledgements

1 The unevenness of biodiversity: corner stone for the large-scale study of life

The astonishing diversity of life on Earth has fascinated scientists for centuries. A study dating back to 15 years ago indeed estimated that around 8.7 million eukaryotic species are currently coexisting on our planet (Mora *et al.* 2011). This number would probably be significantly increased by accounting for the substantial improvements that have occurred in the field of systematics over the recent years, and expanded by orders of magnitude if prokaryotic species were incorporated. Yet, someone travelling to the Sahara desert would probably be less amazed by the surrounding diversity of life than someone travelling to the middle of the Amazonia, one of the most species-rich regions in the world (Raven *et al.* 2020). Another vertiginous data is that around 99.9% of the species that ever existed since the dawn of life on Earth, some 4 billion years ago, has gone extinct. It therefore stands out that, far from a fixed and smooth entity, biodiversity, here understood as species richness, presents a myriad of heterogeneities, that stimulated a vast quantity of research works in which this thesis happens to be nested.

1.1 The heterogeneous organisation of life across space

For more than two centuries, it has been recognised that species are not randomly distributed at the surface of the Earth. In his pioneering “*Essay on the geography of plants*”, regarded as foundational for the science of Biogeography (see Glossary), Alexander von Humboldt emphasised the spatial nature of plant taxonomic assemblages (Humboldt & Bonpland 1807). Later in the 19th century, the scientific community acquired a uniquely integrated perspective on biodiversity fed by years of field observations, giving to the question of the spatial organisation of life on Earth a new dimension. Consequently, it has been proposed to subdivide the world into spatial units sharing similar fauna/flora, sometimes called ‘Homozoic Belts’, ‘Regions’, ‘Realms’, ‘Provinces’, ‘Areas’ or ‘*Gebiet*’ (German word for territory, area) (Burkill 1943). The history will mainly remember the efforts of Alfred R. Wallace in that respect, who relied on the subdivision made by Philip L. Sclater in 1874 based on mammals, as well as his own expertise, to establish a famous contouring on the earth’s ‘Zoogeographic Realms’ (Figure 1.1), two years after (Wallace 1876). Further efforts towards incorporating fossil evidence to better constrain the aforementioned subdivision have to be acknowledged, notably implused by Richard Lydekker (Lydekker 1896). Wallace’s subdivision has maintained in the field since its publication, and only received minor updates in the last decade (Holt *et al.* 2013), illustrating the degree of knowledge advancement of the English naturalist, who lived in a context where Charles Darwin’s famous Theory of Evolution still had a long way to go before being accepted by the broad scientific community.

Beyond its spatial organisation into units of shared taxonomic composition, biodiversity also exhibits a striking latitudinal organisation, in which species richness increases with decreasing latitude, known as the Latitudinal Biodiversity Gradient (LBG). Humboldt already noted that plant



Figure 1.1: Biogeographic map based on faunal similarities within animal assemblages across space ('Zoogeographic Realms') proposed by Wallace (1876).

species richness changed across latitude. According to him, low-latitude tropical assemblages would be essentially made up with a large number of different species per unit of area, each having a relatively low density ("isolated and sparse"). Conversely, higher-latitude temperate or cold assemblages would be characterised by a substantially lower species richness per unit of area, with a single dominant species being over-represented, sort of "excluding [...] heterogeneous species" (Humboldt & Bonpland 1807). LBG raised by Humboldt or Wallace, and first quantified by Fischer (1960). Hillebrand (2004) produced a meta-analysis of 600 different studies and illustrated the consistency of the decrease in species richness with increasing latitude characterising the LBG

Hence, biodiversity is largely heterogeneous across space, raising general questions that have been thoroughly investigated in the past, as to 'Why are there more species in the tropics than in the poles?' 'What constraints the distribution of species on our planet?'

1.2 From peaks to declines: temporally-heterogeneous biodiversity

The aforementioned complex spatial arrangement of life is in fact the legacy of hundreds of millions of years in which species have been fashioned by a wealth of forces – of which we intend to provide an overview in Section 2.2.2. The statement that species experience changes across time, and that these species-level temporal changes eventually scale up to multispecies assemblages making up biodiversity is now commonly accepted, mostly thanks to fossil data.

Being the only source of direct evidence of life in the past, fossils have indeed been of immense value to characterise the temporal dynamics of biodiversity, notably since they irrefutably document the occurrence of extinction in the history of life. It is however interesting to notice that the reasoning that makes us conclude that if, for instance, mammoths are found in the fossil record

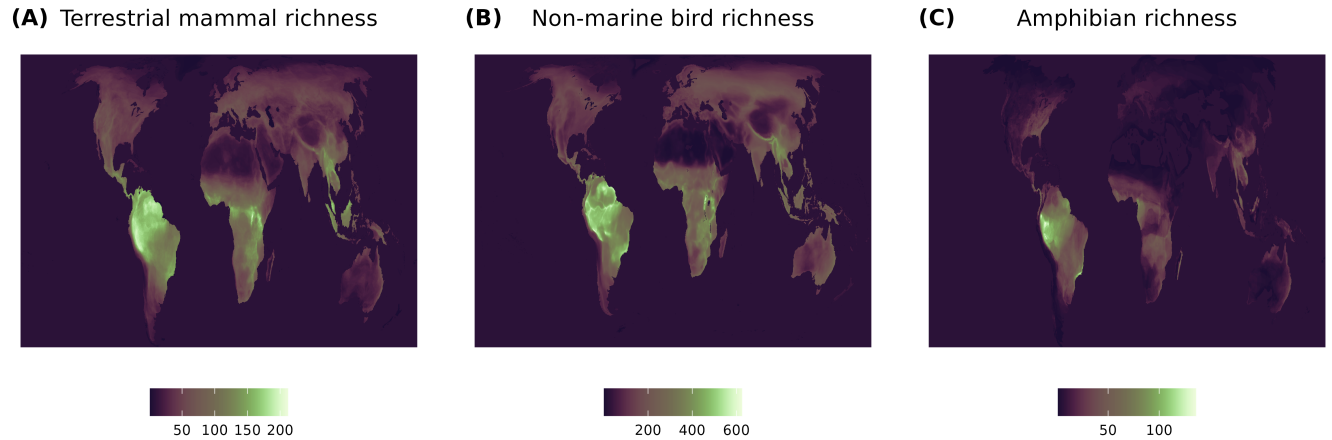


Figure 1.2: Extant land mammal (A), non-marine bird (B) and amphibian (C) species richness at the surface of the globe, expressed in number of species, with a 10 x 10km resolution. Data are from <https://biodiversitymapping.org/>

but not among extant assemblages, it is because they have gone extinct, has not always been as straightforward as it appears to us. Fossils have indeed puzzled eminent figures in the field of biology, such as Jean-Baptiste de Lamarck – to whom we incidentally owe the name ‘biology’. In the early 1800s, Lamarck tentatively formalised a theory where animals would be subject to a complexifying force driving body plans towards higher levels. According to his theory, the (animal) biosphere would thus represent a continuum where species would be unidirectionally connected between each others, which, importantly, leaves no room for extinction – or any sort of species richness variation across time – : every member of Lamarck’s gradual sequence towards increasingly-complex body plans would already exist (Lamarck 1815). However, Lamarck had to cope with the fact that the field of systematic palaeontology was emerging in parallel, notably impulsed by his colleagues from the Muséum National d’Histoire Naturelle de Paris, including Georges Cuvier (Servais *et al.* 2012). Plenty of remains of animals trapped inside the rocks but also manifestly absent from the surface of the (known) world were thus unearthed and described, challenging the Lamarck’s model. To intend to make sense of this apparent inadequacy, Lamarck therefore formulated a famous *ad-hoc* hypothesis: the apparent absence of extant representatives of mammoths does not mean that mammoths have gone extinct. If they indeed locally disappeared from the places where they left fossil remains, they must have found refugia in a region of the globe unexplored by the occidental civilisation.

Markedly opposed to Lamarck’s theory of animal organisation following increasingly complex body plans, Cuvier was maybe one of the earliest scientists to propose that species assemblages could change across time. His rich fieldwork expertise led him to postulate that the history of the globe has been punctuated by abrupt ‘revolution’ events, which can be viewed as (abiotic) changes occurring within a given environment, in turn causing changes in the nature of the species present within this changing environment (Cuvier 1840). Two decades later, John Phillips, an English palaeontologist known as the father of the Phanerozoic eras (Palaeozoic, Mesozoic and Cenozoic) compiled for the first time fossil occurrences of marine taxa from the British lithological record, that he used to produce the first continuous diversity-through-time curve, spanning the entire Phanerozoic (Phillips 1860). His work allowed him to discriminate what he called the Palaeozoic, Mesozoic and Caenozoic ‘lives’. These three ‘lives’ were separated by vast depressions, corresponding to the

Permo–Triassic and Cretaceous–Palaeogene boundaries (Figure 1.3 A), that will be later identified as two amongst the five most violent extinction episodes that marked the history of life, also known as mass extinction crises (see Section 2.2.1). Phillips’ cornerstone compilation thus, for the first time, quantitatively demonstrated the heterogeneity of species richness across geological times.

A century later, the rise of quantitative palaeobiology (see Section 1.4.2) paved the way towards more accurate reconstructions of biodiversity across time. The publication, in the early 1950s, of the first edition of the *Treatise of Invertebrate Paleontology* provided an access to a vast quantity of marine invertebrate fossil data, that could be further used for such an achievement. In that framework, Norman Newell reconstructed diversity curves for about 16 different groups (Newell 1952). Again, the reconstructed curves displayed time-varying patterns. Newell noticed certain similarities in the reconstructed patterns between groups, that he suggested to be evidence for “common causes” underlying their large-scale evolutionary trends. A few decades later, with a higher-resolution curve, Valentine (1969) (Figure 1.3 B) proposed to “explain diversity trends in terms of ecological hierarchy” (Sepkoski 2012), constituting a significant advance towards integrating ecological and evolutionary processes (see Chapter 2).

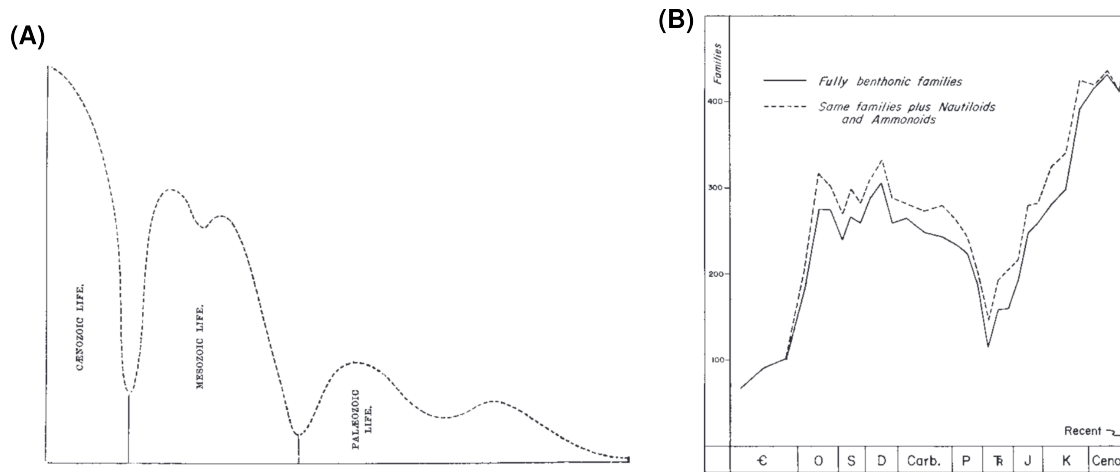


Figure 1.3: Historical fossil-based reconstructions of marine diversity across geological times produced by Phillips (1860), using a marine invertebrate compilation from the British lithological record (A) and by Valentine (1969), using marine invertebrate record from the *Treatise of Invertebrate Paleontology* (B). Note that timescale in (A) has been reversed when compared (B), going back in time from left to right instead of right to left.

However, Valentine’s reconstructions raised some methodological criticism, especially expressed by David M. Raup, one of the leading figures of the ‘palaeobiological revolution’. Raup was indeed concerned about how the fossil record biases may affect Valentine’s reconstruction, importantly stressing that “the increase in the number of marine species across the Phanerozoic may be more apparent than real” (Raup 1972, see Section 1.4.2). Ten years of intense data compilation and methodological labour after, together with J. John Sepkoski Jr. (another major actor of the ‘palaeobiological revolution’) Raup came to the conclusion that marine biodiversity indeed experienced temporal variations, persisting even after accounting for biases inherent to the fossil record (Sepkoski *et al.* 1981). A year after, Raup & Sepkoski (1982) characterised major temporal features in their diversity curve (Figure 1.4), now known as the ‘Big Five mass extinctions’ (that we will develop in further details in Section 2.2.1).

Nowadays, it makes no doubt that biodiversity is temporally heterogeneous. Clades originate, thrive and then possibly decline or even go extinct, sometimes fluctuating in species number in between. A core aim of macroevolution, in which this doctoral work is nested, is thus to elucidate, given a clade, how species diversity changes across time, and the factors underlying these changes (Section 2.2).

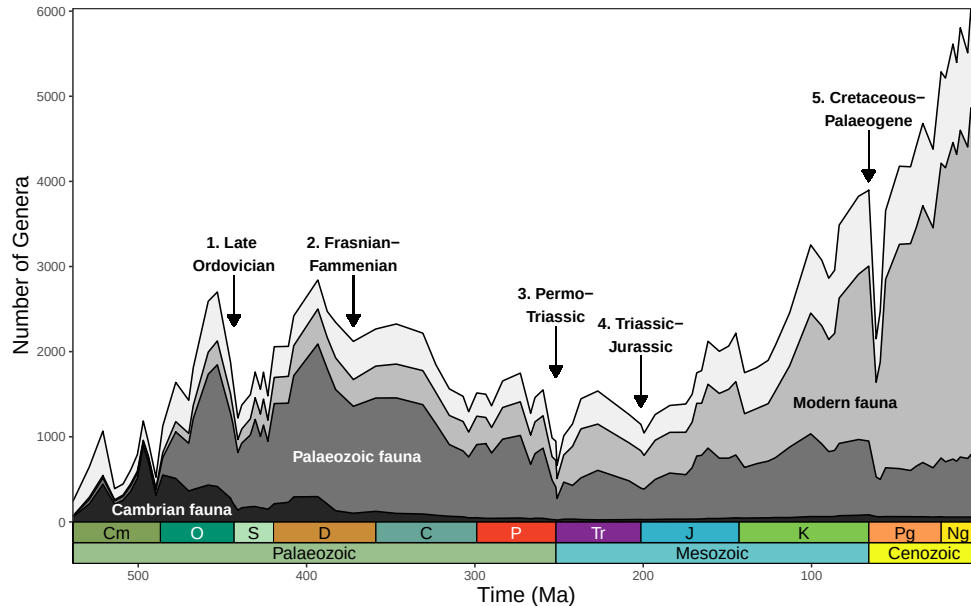


Figure 1.4: Temporal variations in marine animal biodiversity across the Phanerozoic. Colours indicate the three evolutionary faunas discriminated by Sepkoski (1984). Taxa that could not be assigned to any fauna are included in the clearest band. Arrows indicate the “Big Five” mass extinctions, as defined by Raup & Sepkoski (1982). Data are from the genus-level compendium released by Sepkoski (2002), binned across geological stages, accessed and displayed thanks to the R package `sepkoski` (Jones 2022). Geological timescale was added thanks to the `deeptime` R package (Gearty 2023).

1.3 Inter-clade heterogeneity

Another facet of the heterogeneity of biodiversity comes from the fact that some groups harbour an immensely larger species richness than others. This striking feature has intrigued evolutionary biologists since the times of Darwin. In a 1879 letter to Joseph Hooker, the renowned white-bearded English naturalist indeed famously termed the astonishing diversity of flowering plants (Angiosperma) and their apparent rapid burst in the fossil record as an “abominable mystery”. At present, with more than 300,000 described species, angiosperms encompass nearly 90% of the extant land plant diversity.

Over the past few decades, the rise of phylogenetics (Section 3.1.1) permitted to better constrain evolutionary relationships between lineages, thus highlighting inter-clade diversity disequilibria (Figure 1.5). Understanding what drives striking diversity disequilibria between groups, even sometimes closely related, is a central aim of macroevolutionary research.

1.4 Methodological biases arising when studying biodiversity

1.4.1 Biodiversity shortfalls

1.4.2 Propagation of the caveats of the fossil record when studying biodiversity through time

Additional biases emerge as one intends to use fossil data to provide a temporal perspective on biodiversity reconstructions, merely coming from the inherent incompleteness of the fossil record.

Darwin himself recognised the imperfection of the fossil record, comparing the latter to a book written in different dialects with many missing pages, and with only a few lines preserved on the remaining pages Darwin (1859). This conception of the fossil record, too sparse to be accurately used to reconstruct evolutionary processes (“Darwin’s dilemma”, Sepkoski (2012)) was largely shared among the evolutionary biology community, and resulted in the marginalisation of palaeontology from the rest of the community during most of the 20th century. Roughly between 1970 and 1985, a massive change of paradigm occurred as some palaeontologists started to focus on large-scale analyses of evolutionary patterns using quantitative methods, known as the “palaeobiological revolution” Sepkoski (2012).

Early in the history of fossil-based reconstructions of diversity through time, scientists expressed concerns regarding sampling-related biases. In its cornerstone reconstruction, Phillips (1860) had the idea of controlling the richness estimates he obtained per bins with the thickness of the corresponding stratigraphic section. Diversity through time controversy: Raup (1972) and Sepkoski (2012)

Signor & Lipps effect Signor & Lipps (1982)

Close *et al.* (2020)

Thus, studying large-scale biodiversity requires embracing uncertainty!

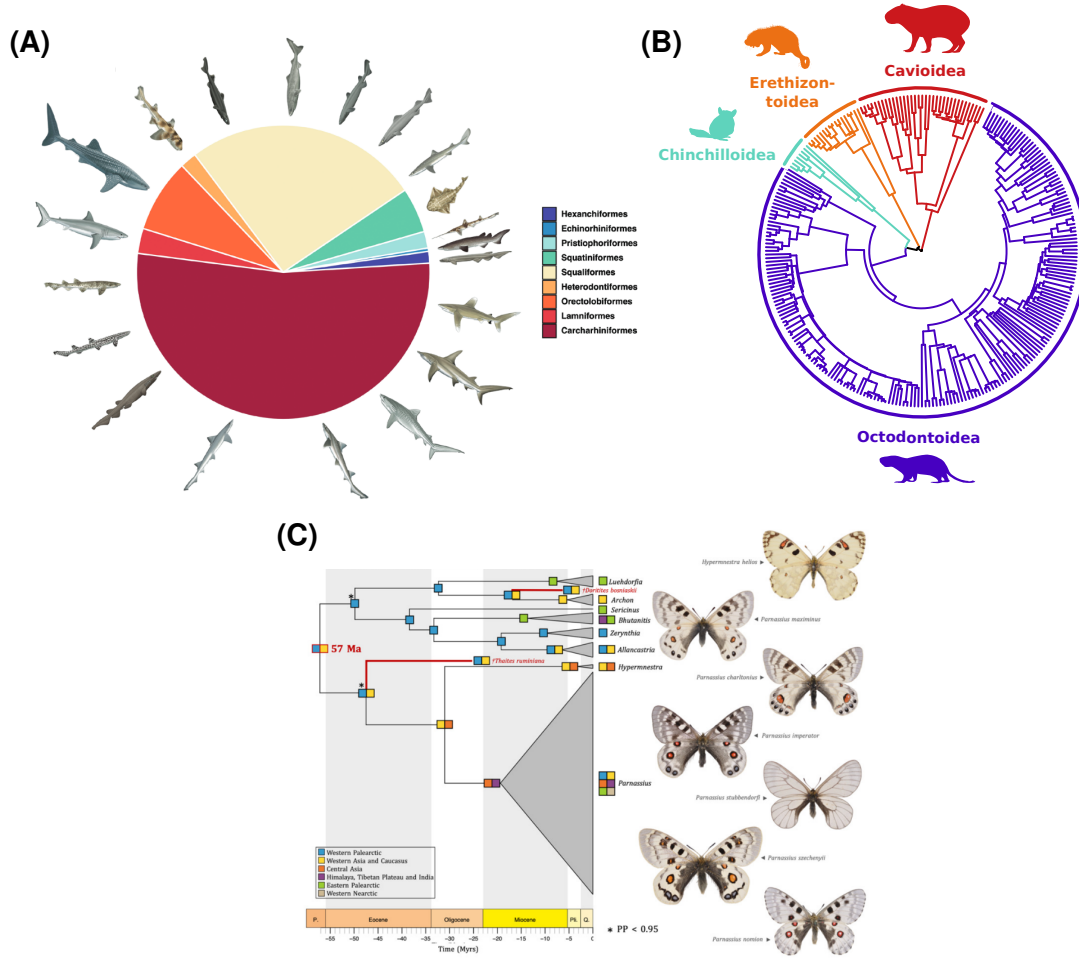


Figure 1.5: Examples of diversity disequilibria in the animal tree of life. **(A)** >50% of the extant shark (Neoselachii) diversity is comprised within the order Carcharhiniformes, whereas other orders (*e.g.*, Echinorhiniformes, Hexanchiformes) are comparatively way less diversified (figure reproduced with permission from Marion *et al.* (2024)). **(B)** ~80% of the extant caviomorph (Rodentia: Hystricognathi) diversity is comprised within the superfamily Octodontoidea, whereas the Chinchilloidea (see *Chapter 2*) only comprises 8 living species. Phylogeny is from VerLife, and silhouettes are from Phylopic (Credits : Michael Tripoli and Gabriela Palomo Muñoz). **(C)** ~90% of the extant diversity of Parnassiinae (Lepidoptera: Papilionidae) is comprised within the genus *Parnassius* (figure reproduced with permission from Hévin & Condamine (2024)).

2 Disentangling biodiversity patterns across space and time: the macro-eco-evolutionary perspective

2.1 Context and terminology

Key concepts and terminology

Speciation, extinction, diversification

Key relating process: biogeography.

2.2 Macroevolution: evolution above the species level

Macroevolution is usually understood as the study of evolutionary patterns occurring above the species level ([Hautmann 2020](#); [Saupe & Myers 2021](#)). Macroevolution has been classically approached under two non-mutually exclusive lenses ([Stanley 1979](#)), either focusing on (1) the diversification process (see Section 2.1 and [Glossary](#)), or (2) the evolutionary dynamics of phenotypes – as an echo to the term ‘diversification’, some authors refer to the process underlying morphological diversity as ‘disparification’ (*e.g.*, [Davis et al. 2016](#); [Nürk et al. 2019](#)). Unfortunately, despite their undeniable appeal, questions related to the dynamics of morphological evolution fall a bit outside the scope of this thesis. Thus, in this manuscript, emphasis will be placed on the tempo and mode of clade diversification.

Just like any other field of evolutionary biology ([Simpson 1944](#)), studying macroevolution is a matter of tempo (*i.e.*, the rate at which macroevolutionary changes occur, namely, speciation, extinction or morphological evolution rates) mode (*i.e.*, the forces underlying these changes) ([Saupe & Myers 2021](#)).

2.2.1 Tempo of Macroevolution: how do rates change across time?

How speciation and extinction rates vary through time mirrors the aforementioned changes in clade diversity across geological times (Section 1.2).

About evolutionary constraints: Raup ([1966](#))

- early acquisition of hovering flight and nectarivory among birds likely promoted the radiation of hummingbirds (Trochilidae) ([Mayr 2004](#))
- early acquisition of echolocation likely favoured the radiation of bats ([Hand et al. 2023](#))

- tooth loss and reappraisal in other diverse forms, believed to have leveraged the morphological constraints imposed by the tribosphenic geometry Vizcaíno (2009).
- flower, along with other traits related to physiology (faster pace of living) promoted angiosperm radiation, Darwin's 'second abominable mystery' (Benton *et al.* 2022; Sauquet *et al.* 2017)

Mass extinctions: Raup & Sepkoski (1982) Raup (1986), Sepkoski (2012) chap. 9

Biodiversity crisis. Parallel with the present-day

2.2.2 Mode of Macroevolution: what drives macroevolutionary rate variations?

It has long been established that biodiversity exhibits marked variations through time, that could not be explained by random fluctuations alone (Sepkoski *et al.* 1981). Since temporal variations of biodiversity can be either understood in terms of pace at which species originate or go extinct, this conclusion suggest that external factors may directly act on the speciation or the extinction processes. Seminal theories have been proposed to relate such environmental dynamics to the pace at which large-scale evolution occurs.

2.2.2.1 Abiotic factors: the 'Court Jester' hypothesis

Despite the relatively recent formulation of the 'Court Jester' model (Barnosky 2001), the idea that abiotic factors may affect diversification rates has been in the minds for a long time.

Palaeogeography, climate

2.2.2.2 Biotic factors: the 'Red Queen' hypothesis

A turning point in the appreciation of the macroevolutionary drivers underlying clade evolution arose from the ecological theory, in the second half of the 20th century, notably coming from the formalisation of the concept of ecological niche.

Concept of 'competitive exclusion' first developed in 1936 with the formalisation of the Lotka-Volterra model, and experiments using *Paramecium* (see Diniz-Filho (2023)).

Concept of 'character displacement' Pfenning & Pfenning (2009)

2.2.2.3 Traits

The idea that phenotypical traits can influence diversification has been around since the formulation of the concept of adaptive radiation, where the emergence of a phenotypical feature ('key innovation') confers to a clade the ability ("ecological opportunity") to radiate within a previously unexploited portion of the environment (see Simpson 1944). Traits may also mediate biotic interactions.

2.3 Macro-eco-evolutionary understanding of the spatial organisation of life: insights from the Latitudinal Biodiversity Gradient

The uneven species richness across the globe that we outlined in Section 1.1, and notably, the striking pattern of increasing species richness with decreasing latitude famously known as the ‘Latitudinal Biodiversity Gradient’ (LBG), can be understood under a macroevolutionary perspective involving ecological processes. Several hypotheses (*reviewed in Pontarp et al. 2019*) have been proposed to explain such a pattern, some of which imply the interplay between diversification rates and tropical environmental conditions (*Tietje et al. 2022*).

In the footsteps of Stebbins (1974), the ‘cradle’ hypothesis posits that tropical lineages would have larger speciation rates than extra-tropical lineages. Higher speciation rates for tropical lineages may be predicted from tropic’s greater environmental heterogeneity (*Stein et al. 2014*), increasing available niche space. Conversely, one can expect speciation rates to be lower outside the tropics, where climate seasonality is more pronounced. In addition, the ‘Metabolic Theory of Ecology’ (*Brown et al. 2004*) can also predict that tropical lineages would have larger speciation rates. According to the latter, the higher energy availability for the tropics (higher temperature) would locally increase tropical species’ metabolic rate (*Allen et al. 2002*), yielding larger mutation, and thus, speciation rates (*Allen et al. 2006; Lanfear et al. 2010*). Also, as summarised by *Mittelbach et al. (2007)*, tropical regions’ greater species richness has been suggested to enhance the importance of biotic interactions (*Schemske et al. 2009*). Consequently, since the ‘Red Queen’ hypothesis (Section 2.2.2) postulates that biotic interactions would enhance speciation, this would establish a positive feedback loop within the tropics, where increasing diversity would yield increasing speciation rates. This ‘cradle’ hypothesis is often contrasted with the ‘museum’ hypothesis, instead placing the emphasis on the fact that tropical lineages would have lower extinction rates. The ‘museum’ hypothesis may be a consequence of the ‘tropical stability’ hypothesis, assuming that the tropics have remained climatically stable across geological times, and that this stability would have made tropical lineages less prone to extinction (*Wallace 1878*). In addition, it might be expected that the aforementioned environmental heterogeneity offered by the tropics could buffer tropical lineages against eventual environmental changes.

A consensus emerged from late 2000s–2010s studies testing the extent to which diversification rates could explain the observed LBG, where the tropics may actually both act as a cradle and a museum of diversity (*e.g., Jablonski et al. 2006; Marshall 2006; Mittelbach et al. 2007; Moreau & Bell 2013; Rolland et al. 2014*). This view has been contrastingly supported in the recent literature. Fossil-based analyses of the South American Mammal fossil record supported this view, with tropical taxa showing elevated speciation rates and reduced extinction rates (*Buffan et al. 2025*). Nevertheless, joint analyses of fossil and phylogenetic data for several tetrapod clades, using cutting-edge methodological frameworks, provided support for elevated speciation rate outside the tropics as a likely mechanism for the LBG, thus instead favouring the ‘museum’ hypothesis (*Meseguer & Condamine 2020; Quintero et al. 2023*). Conversely, a recent study focusing on a massive sample of the seed plant diversity emphasised that no causal relationship could be established between diversification rates and species richness (*Tietje et al. 2022*). Given these elements, it seems like the study model, the primary source of data (*e.g., fossils vs. dated phylogenies*) and the methodology used to estimate diversification rates (see Section 3.1) affects upstream interpretations of the interplay between diversification rates and spatial patterns of species richness.

However, the notions of ‘cradle’ and ‘museum’ of biodiversity have recently been challenged by *Vasconcelos et al. (2022)*, emphasising a number of limitations for the way they are often consid-

ered in the literature. Among them, as several factors driving speciation, such as climatic stability/heterogeneity, may also drive extinction, they argue that it would sometimes make more sense to consider turnover rates (see [Glossary](#)) rather than speciation and extinction rates individually. Also, they point out that extant centres of biodiversity with a great concentration of ‘basal’ lineages are not necessarily centres of origin, thus challenging both the way previous works actually interpreted the notion of museum and cradle of diversity. The latter nuance echoes previous works of the same authors emphasising the need to include hidden states in models of character evolution, marking a turning point in macroevolution (*e.g.*, [Beaulieu & O’Meara 2016](#)).

Alternative explanations of the LBG not involving diversification rates have also been proposed. These either refer to species’ ecological limits (*e.g.*, carrying capacity, size of the available area, palaeogeography) or time left for species to diversify (assuming tropical stability would have left more time for species to diversify within than outside the tropics, and attested by fossil and phylogenetic evidence of lineage age distribution in and outside the tropics) ([Pontarp *et al.* 2019](#)). As stressed by [Mittelbach *et al.* \(2007\)](#), these three classes of hypotheses (diversification-, ecological- or time-dependent) are not necessarily mutually exclusive. Conversely, a fourth set of hypotheses emphasised the likely implication of pure contingency in shaping the LBG, notably through the use of null models ([Arita & Vázquez-Domínguez 2008](#)). Recently insights from eco-evolutionary simulations (see [Section 3.2](#)) evidenced that null models – where the modelled eco-evolutionary processes do not rely on any environmental constraint – could indeed match empirical LBG patterns, although generally worse at explaining empirical distribution data than model incorporating mechanistic constraints on species, function of their biotic and abiotic environment ([Hagen *et al.* 2021](#)).

3 Analytical tools to study large-scale biodiversity patterns

Means to study diversification

3.1 Macroevolutionary approaches to study how clades thrive and go extinct

Review from Benton ([2015](#))

3.1.1 Phylogeny-based

3.1.1.1 Overview

What is a phylogeny?

Phylogeny-based methods

3.1.1.2 Limitations

3.1.2 Fossil occurrence-based

3.1.2.1 Overview

DivDyn, PyRate

3.1.2.2 Limitations

3.1.3 Towards integrating multiple sources of data

TED

3.2 Integrating ecological processes with evolutionary forces: mechanistic eco-evolutionary simulations

4 A strange biodiversity on a strange land : the case of mammals in South America

South America has attracted the interest of natural scientists since the times of Humboldt, Darwin or Wallace.

In 1876, Wallace wrote that the Neotropical region, encompassing South America, Caribbean Islands and tropical North America “possesses more peculiar families of vertebrates and genera of birds and mammalia than any other region” ([Wallace 1876](#)).

Introduce Wallaces’ zoogeographical subregions for the Neotropics (Mexican, Brazilian, Antillean, Chilean)?

4.1 South America across the Cenozoic: geographical and environmental context

Start from SA ecoregions in the present day (plot map, and overlay with diversity of clades).

Palaeoenvironmental evolution: summarise & update Hoorn *et al.* ([2010](#))

Continental island framework

4.1.1 Amazonia: a unique and dynamic tropical ecoregion

4.1.2 The rise of the Andes : emergence of one of the largest and highest mountain chains, and its consequences on biodiversity

4.1.3 Intermediate flooding events

4.1.4 South America as a playground for evolutionary biologists

4.2 ‘Splendid isolation’: the captivating history of South American mammals across the Cenozoic

“the southern tree of life, its own tree and independent of the one born and raised in the opposite hemisphere” (Francisco Moreno, El Museo de La Plata: Rápida ojeada sobre su fundación y desarrollo.” Revista del Museo de La Plata 1 (1890–1891): 27–55)

Due to its astonishing diversity mostly unknown to Europeans at the time South America has almost acted as an ‘obligate step’

4.2.1 History of mammal palaeontology in South America: a brief overview

Paul Gervais?

South American mammal palaeontology sadly started as a colonisation by-product, as European explorators plundered Patagonian soils, intrigued by skeletal remains trapped in the rocks. In the late 18th century, some of the South American fossils (essentially from the Southern South American plains) brought back to Europe provoked curiosity among the leading figures of the emerging field of comparative anatomy. As a result, in 1796, Georges Cuvier described the famous *Megatherium americanum*, and proposed this six-meter-beast to be a sloth. A few decades later, in 1836, Charles Darwin, back from his five-years journey on board the HMS Beagle building a geological and naturalistic inventory in South America, brought back to England a wealth of fossil material (in particular from the region of Punta Alta, Patagonia). These fossil remains were given to the renowned palaeontologist Richard Owen, who described, in 1837, many emblematic fossil mammals from South America, including the car-sized armadillo *Glyptodon*, the giant sloth *Myodon*, the giant camel-like native ungulate *Macrauchenia* or the rhino-sized native ungulate *Toxodon*.

The second half of the 19th century marked the beginning of South American mammal palaeontology made by scientists from South America, notably with the thorough and cornerstone labour of Florentino Ameghino. This Argentinian palaeontologist, born in Italy, produced a remarkable quantity of work relying on fossil material collected by his brother, Carlos Ameghino, who carried out most of the field expeditions that fed Florentino Ameghino's collections. Florentino Ameghino's written productions are regarded as essential building-blocks for South American palaeomammalogy. These are probably best exemplified by the foundational monography "*Contribución al conocimiento de los mamíferos fósiles de la República Argentina*" (1889), where several hundreds of taxa from the Argentinian fossil record are described, many of them still considered as valid species. The beginning of the 1910s saw the rise of another greatly recognised Argentinian figure of South American mammal palaeontology: Lucas Kraglievich. Kraglievich conducted key field expeditions in Patagonia. The material he collected led him to produce major contributions to the knowledge of the evolution of xenarthrans, caviomorphs and native ungulates.

South American mammal palaeontology long remained essentially constrained to Argentina, as it shows on the map of the distribution of fossil mammal-bearing localities on the continent (see Section 4.3 and Figure X).

Apart from Argentina, Brazil in the 1950s (Carlos de Paula Couto).

Simpson

Check Manias (2023)

Figure with portraits of all the names mentioned?

Map SA with Punta Alta + bones in question?

Simpson (1980) wrote that South American mammal assemblages across the Cenozoic were best summarised by three successive temporal strata.

4.2.2 First stratum: endemic radiations (late Cretaceous–middle Eocene)

Xenarthrans, metatherians, SANUs

4.2.3 Second stratum: Palaeogene migrant arrival and diversification (late Eocene–Late Miocene)

4.2.4 Third stratum: North American migrant arrival, massive faunal disruption, and establishment of the extant mammal biodiversity across the Neotropics (latest Miocene–Present)

Tropical North America: “the portion of the North American continent which belongs zoologically to South America” ([Wallace 1876](#)).

4.3 Challenges for South American mammal palaeontology

Striking unevenness of the fossil record Ugarte *et al.* ([2025](#)). This unevenness is spatial (map of occ) and temporal (important temporal gaps persist in the fossil record of SAM, display SALMA scale). Difficulty to sample. Colonisation palaeontology.

5 Objectives of this doctoral work

5.1 Deciphering the dynamics of diversification of South American mammals and their likely drivers across the Cenozoic

Can we explain the emergence of key phenotypical features (hypsodonty, body size)?

5.1.1 Elucidating the impact of the Eocene–Oligocene Transition on the macroevolutionary dynamics of South American mammals

5.1.2 Teasing apart the legacy of the Neogene continent-scale environmental upheavals on the macroevolution of SAM

5.1.3 Investigating what drove the emergence and demise of the largest rodents ever fooling the Earth: Chinchilloidea

5.2 Assessing the role of palaeoclimate in the emergence of the asymmetry behind the Great American Biotic Interchange

Chapter 1. Deciphering the trends and drivers of South American mammal large-scale evolution across the Cenozoic

Chapter 2.1. The fate of South America's endemic mammalian fauna in response to the most dramatic Cenozoic climate disruption

Chapter 2.2. Continent-scale insights into the last 23 million years of South American mammalian evolution

Chapter 2. A matter of size: what underpinned the rise and demise of the largest rodents that ever lived?

Chapter 3. Palaeoclimate alone fails to explain the asymmetry of the Great American Biotic Interchange

Discussion

Limitations

Lack of time-continuous proxy for precipitations.

Lack of well-constrained phylogenetic framework, as many of the endemic Cenozoic South American mammal fauna has gone extinct

Terminology, definitions and other macroevolutionary enigmas

Defining a species?

Inferring diversity/diversification patterns from the fossil record: Signor & Lipps effect Signor & Lipps ([1982](#))

Perspectives

An almost closed system: founding migrations in the Palaeogene history of South American mammals

Hystricognath rodents & Platyrrhine primates. Introduce the debate. We lack fossil evidence for the earliest comers, oldest occurrences in Western Amazonia whereas most likely come from Afro-Asian ancestors, so should have arrived by the East. Polyphyletic/several crossing events? Summarised in Marivaux *et al.* ([2023](#))

Pino *et al.* ([n.d.](#)) for migrations and impact on SA diversity

There is still some work to do to accurately elucidate continent-scale macroevolutionary dynamics of Caviomorph rodents (others than Chinchilloids, a contribution of this thesis): set the scene for GIGARAT

Sustainable late Cenozoic connection to North America: what drove the asymmetry of the GABI?

Set the scene for IMBALANCE

How about neutral theory?

Future directions towards an improved understanding of the emergence and dynamics of South American ecosystems

DIACHRONANDES, PHENOMENA

Concluding remarks

Glossary

This section intends to provide simple definitions to key notions and concepts to which this thesis refers. Note that some are complex and could be the subject of entire research works. Thus, these definitions do not aim to be as exhaustive as possible, but rather, to improve the reader's understanding of this manuscript.

Adaptive radiation. Rapid *diversification* of a group taking advantage of a previously unexploited ecological dimension of the environment, generally spurred by a *key innovation*. This concept has been proposed by Simpson (1944).

Angiosperm Terrestrial Revolution (ATR). Suite of major global-scale ecosystem reshaping that followed the radiation of flowering plants (angiosperms) in the Early Cretaceous (*ca.* 140 Ma). The term was introduced by Benton *et al.* (2022).

Biogeography. Discipline aiming to describe and understand the spatial distribution of species.

Clade. Given a multi-species sample, a clade is an exhaustive subsample of all species sharing a common ancestry. Clades are referred to as *monophyletic*.

Diversification. The diversification process summarises the net evolutionary changes in a *clade* across time, that is, the gains (modelled by the *speciation* process) minus the losses (modelled by the *extinction* process). As such, it represents the process generating clade's diversity. The intensity of the diversification process (called diversification rate, often noted $d(t)$ for a given time t) is thus the difference between the rates of speciation and extinction, leading to $d(t) = \lambda(t) - \mu(t)$.

Ecological niche. The concept of ecological niche may be understood in different ways. In macroecology, the niche is more often viewed from a perspective of the species' requirements, therefore following the framework proposed by Grinnell (1917) and later Hutchinson (1957). Considering a landscape made up with n ecological variables (*e.g.*, climate, topography, abundance of other species, abundance of trophic resources, ...), the ecological niche of a given species theoretically represents the portion of the landscape where the species occurs. As such, it can be viewed as an n -hyperpolygon encompassing the species' tolerance range following each variable (*e.g.*, 15–25°C, 0–1000m, ...). Other definitions exist, such as the Eltonian niche Elton (1927), more centered on species' impact on the environment.

Extinction. Process of disappearance of all representatives of a given lineage. In *macroevolutionary modelling*, extinction rates (expressed in event per unit of time) are often noted $\mu(t)$.

Latitudinal Biodiversity Gradient (LBG). Pattern of increasing species diversity with decreasing latitude (*i.e.*, within a given unit of the globe's surface, the number of species within it increases as we get closer to the equator), found in a wide range of group.

Speciation. Origination of a new species from an ancestral lineage. There are several known types of speciation. Speciation is called allopatric when a physical barrier (*e.g.*, a mountain, a river) subdivides an ancestral population into several sub-populations, thus interrupting gene flow and driving to the split in as many species as initial subsets by divergence accumulation. Conversely,

a the barrier to gene flow arises within a population where individuals remain in connexion (*e.g.*, behavioural barrier), people refer to *sympatric* speciation. When speciation arises from a founder effect, *i.e.*, colonisation of a new area (*e.g.*, an island), one refers to *peripatry*. A particular case of peripatry, when the colonised area is adjacent to the range of the initial species, is called *parapatry*. Given these several modes, modelling speciation can be done in different ways, defined in this glossary (*cladogenesis*, *anagenesis*, *budding*). In *macroevolutionary modelling*, speciation rates through time, expressed in number of events per unit of time, are often noted $\lambda(t)$

Trait.

Literature cited

- Allen, A.P., Brown, J.H. & Gillooly, J.F. (2002). [Global biodiversity, biochemical kinetics, and the energetic-equivalence rule](#). *Science*, 297, 1545–1548.
- Allen, A.P., Gillooly, J.F., Savage, V.M. & Brown, J.H. (2006). [Kinetic effects of temperature on rates of genetic divergence and speciation](#). *Proceedings of the National Academy of Sciences*, 103, 9130–9135.
- Arita, H.T. & Vázquez-Domínguez, E. (2008). [The tropics: cradle, museum or casino? A dynamic null model for latitudinal gradients of species diversity](#). *Ecology Letters*, 11, 653–663.
- Barnosky, A.D. (2001). [Distinguishing the effects of the red queen and court jester on miocene mammal evolution in the northern rocky mountains](#). *Journal of Vertebrate Paleontology*, 21, 172–185.
- Beaulieu, J.M. & O'Meara, B.C. (2016). [Detecting hidden diversification shifts in models of trait-dependent speciation and extinction](#). *Systematic Biology*, 65, 583–601.
- Benton, M.J. (2015). [Exploring macroevolution using modern and fossil data](#). *Proceedings of the Royal Society B: Biological Sciences*, 282, 20150569.
- Benton, M.J., Wilf, P. & Sauquet, H. (2022). [The Angiosperm Terrestrial Revolution and the origins of modern biodiversity](#). *New Phytologist*, 233, 2017–2035.
- Brown, J.H., Gillooly, J.F., Allen, A.P., Savage, V.M. & West, G.B. (2004). [Toward a Metabolic Theory of Ecology](#). *Ecology*, 85, 1771–1789.
- Buffan, L., Condamine, F.L., Stutz, N.S., Pujos, F., Antoine, P.-O. & Marivaux, L. (2025). [The fate of south america's endemic mammalian fauna in response to the most dramatic cenozoic climate disruption](#). *Proceedings of the National Academy of Sciences*, 122, e2419520122.
- Burkill, I.H. (1943). [The biogeographic division of the indo-australian archipelago. 2. A history of the divisions which have been proposed](#). *Proceedings Linnean Society London*, 154, 127–138.
- Close, R.A., Benson, R.B.J., Alroy, J., Carrano, M.T., Cleary, T.J., Dunne, E.M., *et al.* (2020). [The apparent exponential radiation of phanerozoic land vertebrates is an artefact of spatial sampling biases](#). *Proceedings of the Royal Society B: Biological Sciences*, 287, 20200372.
- Cuvier, G. (1840). *Discours sur les révolutions de la surface du globe*.
- Darwin, C. (1859). *On the origin of species by means of natural selection, or the preservation of favoured races in the struggle for life*. John Murray. London.
- Davis, A.M., Unmack, P.J., Vari, R.P. & Betancur-R., R. (2016). [Herbivory Promotes Dental Disparification and Macroevolutionary Dynamics in Grunters \(Teleostei: Terapontidae\), a Freshwater Adaptive Radiation](#). *The American Naturalist*, 187, 320–333.
- Diniz-Filho, J.A.F. (2023). *The macroecological perspective*. Springer Cham.
- Elton, C. (1927). *Animal ecology*. Sidgwick and Jackson. London.
- Fischer, A.G. (1960). [Latitudinal variations in organic diversity](#). *Evolution*, 14, 64–81.
- Gearty, W. (2023). [Deeptime: Plotting tools for anyone working in deep time. R package version 1.0.1](#).
- Grinnell, J. (1917). [The niche-relationships of the california thrasher](#). *The Auk*, 34, 427–433.
- Hagen, O., Flück, B., Fopp, F., Cabral, J.S., Hartig, F., Pontarp, M., *et al.* (2021). [gen3sis: A general engine for eco-evolutionary simulations of the processes that shape Earth's biodiversity](#).

- PLOS Biology*, 19, e3001340.
- Hand, S.J., Maugoust, J., Beck, R.M.D. & Orliac, M.J. (2023). [A 50-million-year-old, three-dimensionally preserved bat skull supports an early origin for modern echolocation.](#) *Current Biology*, 33, 4624–4640.e21.
- Hautmann, M. (2020). [What is macroevolution?](#) *Palaeontology*, 63, 1–11.
- Hévin, N.M.-C. & Condamine, F.L. (2024). [Bayesian Total-Evidence Dating Reshapes the Age and Historical Biogeography of Parnassiinae.](#) *Bulletin of the Society of Systematic Biologists*, 3.
- Hillebrand, H. (2004). On the generality of the latitudinal diversity gradient. *The American Naturalist*, 163, 192211.
- Holt, B.G., Lessard, J.-P., Borregaard, M.K., Fritz, S.A., Araújo, M.B., Dimitrov, D., *et al.* (2013). [An update of wallace’s zoogeographic regions of the world.](#) *Science*, 339, 74–78.
- Hoorn, C., Wesselingh, F.P., Steege, H. ter, Bermudez, M.A., Mora, A., Sevink, J., *et al.* (2010). [Amazonia through time: Andean uplift, climate change, landscape evolution, and biodiversity.](#) *Science*, 330, 927–931.
- Humboldt, A. von & Bonpland, A. (1807). *Essai sur la géographie des plantes; accompagne d’un tableau physique des régions équinoxiales*. Paris: Fr. Schoell.
- Hutchinson, G.E. (1957). Cold spring harb symp quant biol. pp. 415–427.
- Jablonski, D., Roy, K. & Valentine, J.W. (2006). [Out of the tropics: Evolutionary dynamics of the latitudinal diversity gradient.](#) *Science*, 314, 102–106.
- Jones, L.A. (2022). *Sepkoski: Sepkoski’s fossil marine animal genera compendium*.
- Lamarck, J.-B.P.A. de M. de. (1815). *Histoire naturelle des animaux sans vertèbres*. Paris.
- Lanfear, R., Ho, S.Y.W., Love, D. & Bromham, L. (2010). [Mutation rate is linked to diversification in birds.](#) *Proceedings of the National Academy of Sciences*, 107, 20423–20428.
- Lydekker, R. (1896). *A geographical history of mammals*. Cambridge University Press.
- Manias, C. (2023). *The age of mammals: Nature, development, and paleontology in the long nineteenth century*. University of Pittsburgh Press.
- Marion, A.F.P., Condamine, F.L. & Guinot, G. (2024). [Sequential trait evolution did not drive deep-time diversification in sharks.](#) *Evolution*, qpae070.
- Marivaux, L., Negri, F.R., Antoine, P.-O., Stutz, N.S., Condamine, F.L., Kerber, L., *et al.* (2023). [An eosimiid primate of south asian affinities in the paleogene of western amazonia and the origin of new world monkeys.](#) *Proceedings of the National Academy of Sciences*, 120, e2301338120.
- Marshall, C.R. (2006). [Fossil record reveals tropics as cradle and museum.](#) *Science*, 314, 66–67.
- Mayr, G. (2004). [Old world fossil record of modern-type hummingbirds.](#) *Science*, 304, 861–864.
- Meseguer, A.S. & Condamine, F.L. (2020). [Ancient tropical extinctions at high latitudes contributed to the latitudinal diversity gradient.](#) *Evolution*, 74, 1966–1987.
- Mittelbach, G.G., Schemske, D.W., Cornell, H.V., Allen, A.P., Brown, J.M., Bush, M.B., *et al.* (2007). [Evolution and the latitudinal diversity gradient: speciation, extinction and biogeography.](#) *Ecology Letters*, 10, 315–331.
- Mora, C., Tittensor, D.P., Adl, S., Simpson, A.G.B. & Worm, B. (2011). [How Many Species Are There on Earth and in the Ocean?](#) *PLOS Biology*, 9, e1001127.
- Moreau, C.S. & Bell, C.D. (2013). [Testing the museum versus cradle tropical biological diversity hypothesis: Phylogeny, diversification, and ancestral biogeographic range evolution of the ants.](#) *Evolution*, 67, 2240–2257.
- Newell, N.D. (1952). [Periodicity in invertebrate evolution.](#) *Journal of Paleontology*, 26, 371–385.
- Nürk, N.M., Atchison, G.W. & Hughes, C.E. (2019). [Island woodiness underpins accelerated disparification in plant radiations.](#) *New Phytologist*, 224, 518–531.
- Pfenning, K.S. & Pfenning, D.W. (2009). [Character Displacement: Ecological And Reproductive Responses To A Common Evolutionary Problem.](#) *The Quarterly Review of Biology*, 84.

- Phillips, J. (1860). *Life on the earth; its origin and succession*. MacMillan; co, Cambridge; London.
- Pino, K., Rodriguez-Serrano, E., Carrillo, J.D., Cooper, R.B. & Silvestro, D. (n.d.). [Splendid isolation with migration: Diversity dynamics of South American mammals](#).
- Pontarp, M., Bunnefeld, L., Cabral, J.S., Etienne, R.S., Fritz, S.A., Gillespie, R., *et al.* (2019). [The latitudinal diversity gradient: Novel understanding through mechanistic eco-evolutionary models](#). *Trends in Ecology & Evolution*, 34, 211–223.
- Quintero, I., Landis, M.J., Jetz, W. & Morlon, H. (2023). [The build-up of the present-day tropical diversity of tetrapods](#). *Proceedings of the National Academy of Sciences*, 120, e2220672120.
- Raup, D.M. (1966). [Geometric analysis of shell coiling: General problems](#). *Journal of Paleontology*, 40, 1178–1190.
- Raup, D.M. (1972). [Taxonomic diversity during the phanerozoic](#). *Science*, 177, 1065–1071.
- Raup, D.M. (1986). [Biological extinction in earth history](#). *Science*, 231, 1528–1533.
- Raup, D.M. & Sepkoski, J.J. (1982). [Mass extinctions in the marine fossil record](#). *Science*, 215, 1501–1503.
- Raven, P.H., Gereau, R.E., Phillipson, P.B., Chatelain, C., Jenkins, C.N. & Ulloa Ulloa, C. (2020). [The distribution of biodiversity richness in the tropics](#). *Science Advances*, 6, eabc6228.
- Rolland, J., Condamine, F.L., Jiguet, F. & Morlon, H. (2014). [Faster Speciation and Reduced Extinction in the Tropics Contribute to the Mammalian Latitudinal Diversity Gradient](#). *PLOS Biology*, 12, e1001775.
- Saupe, E.E. & Myers, C.E. (2021). [Macroevolution](#). Springer, Cham, pp. 149–167.
- Sauquet, H., Balthazar, M. von, Magallón, S., Doyle, J.A., Endress, P.K., Bailes, E.J., *et al.* (2017). [The ancestral flower of angiosperms and its early diversification](#). *Nature Communications*, 8, 16047.
- Schemske, D.W., Mittelbach, G.G., Cornell, H.V., Sobel, J.M. & Roy, K. (2009). [Is There a Latitudinal Gradient in the Importance of Biotic Interactions?](#) *Annual Review of Ecology, Evolution, and Systematics*, 40, 245–269.
- Sepkoski, D. (2012). [Rereading the Fossil Record: The Growth of Paleobiology as an Evolutionary Discipline](#). University of Chicago Press, Chicago, IL.
- Sepkoski, J.J. (1984). [A kinetic model of Phanerozoic taxonomic diversity. III. Post-Paleozoic families and mass extinctions](#). *Paleobiology*, 10, 246–267.
- Sepkoski, J.J. (2002). [A compendium of fossil marine animal genera](#). *Bulletins of American Paleontology*, 363, 1–560.
- Sepkoski, J.J., Bambach, R.K., Raup, D.M. & Valentine, J.W. (1981). [Phanerozoic marine diversity and the fossil record](#). *Nature*, 293, 435–437.
- Servais, T., Antoine, P.-O., Danielan, T., Lefebvre, B. & Meyer-Berthaud, B. (2012). Paleontology in france: 200 years in the footsteps of cuvier and lamarck. *Palaeontologia Electronica*, 15.
- Signor, P.W., I. & Lipps, J.H. (1982). [Sampling bias, gradual extinction patterns and catastrophes in the fossil record](#). *Geological Society of America Special papers*, 0.
- Simpson, G.G. (1944). *Tempo and mode in evolution*. Columbia University Press.
- Simpson, G.G. (1980). *Splendid isolation : The curious history of south american mammals*. Yale University Press, New Haven.
- Stanley, S.M. (1979). *Macroevolution: Pattern and process*. W. H. Freeman, San Francisco.
- Stebbins, G.L. (1974). Trends of angiosperm phylogeny - the ecological basis of diversity. Cambridge, Mass, pp. 156–170.
- Stein, A., Gerstner, K. & Kreft, H. (2014). [Environmental heterogeneity as a universal driver of species richness across taxa, biomes and spatial scales](#). *Ecology Letters*, 17, 866–880.
- Tietje, M., Antonelli, A., Baker, W.J., Govaerts, R., Smith, S.A. & Eiserhardt, W.L. (2022). [Global variation in diversification rate and species richness are unlinked in plants](#). *Proceedings of the*

- National Academy of Sciences*, 119, e2120662119.
- Ugarte, P.D. de S., Nascimento, J.C.S. & Pires, M.M. (2025). [Spatiotemporal variability in the south american mammalian fossil record and its impact on macroevolutionary inference](#). *Frontiers in Mammal Science*, 3.
- Valentine, J.W. (1969). Patterns of taxonomic and ecological structure of the shelf benthos during phanerozoic time. *Palaeontology*, 12, 684–709.
- Vasconcelos, T., O'Meara, B.C. & Beaulieu, J.M. (2022). [Retiring “cradles” and “museums” of biodiversity](#). *The American Naturalist*, 199, 194–205.
- Vizcaíno, S.F. (2009). [The teeth of the “toothless”: novelties and key innovations in the evolution of xenarthrans \(Mammalia, Xenarthra\)](#). *Paleobiology*, 35, 343–366.
- Wallace, A.R. (1876). *The geographical distribution of animals. With a study of the relations of living and extinct faunas as elucidating the past changes of the earth's surface*. Harper; brothers, New York.
- Wallace, A.R. (1878). *Tropical Nature, and Other Essays*. Macmillan; Company.